Ex.No. 13

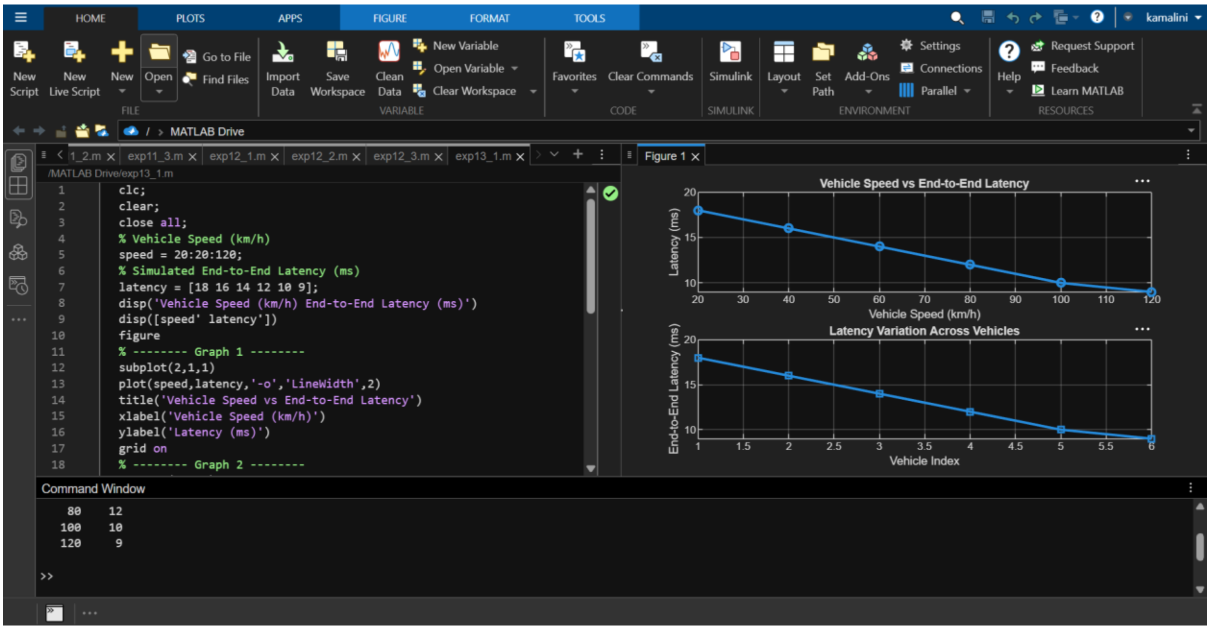
V2X Latency Comparison Date

Objective 1 :

To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.

CODE:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```



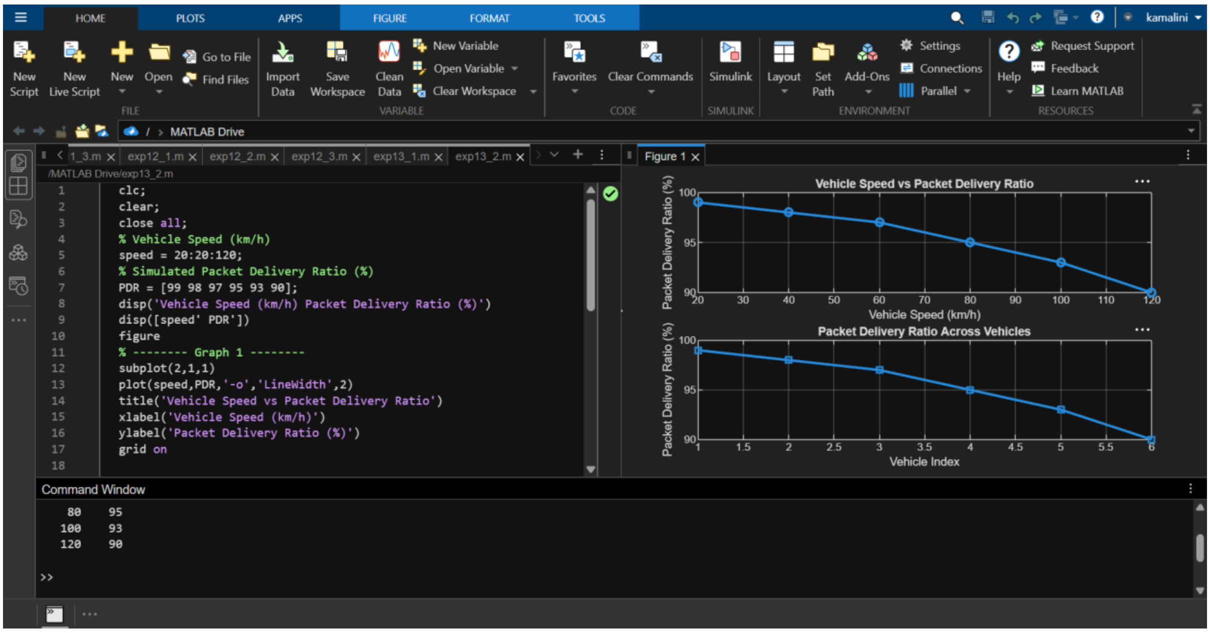
Objective 2 :

To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.

CODE :

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];
disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')
disp([speed' PDR'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,PDR,'-o','LineWidth',2)
title('Vehicle Speed vs Packet Delivery Ratio')
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Packet Delivery Ratio (%)')

grid on


% -------- Graph 2 --------

subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')

grid on
```



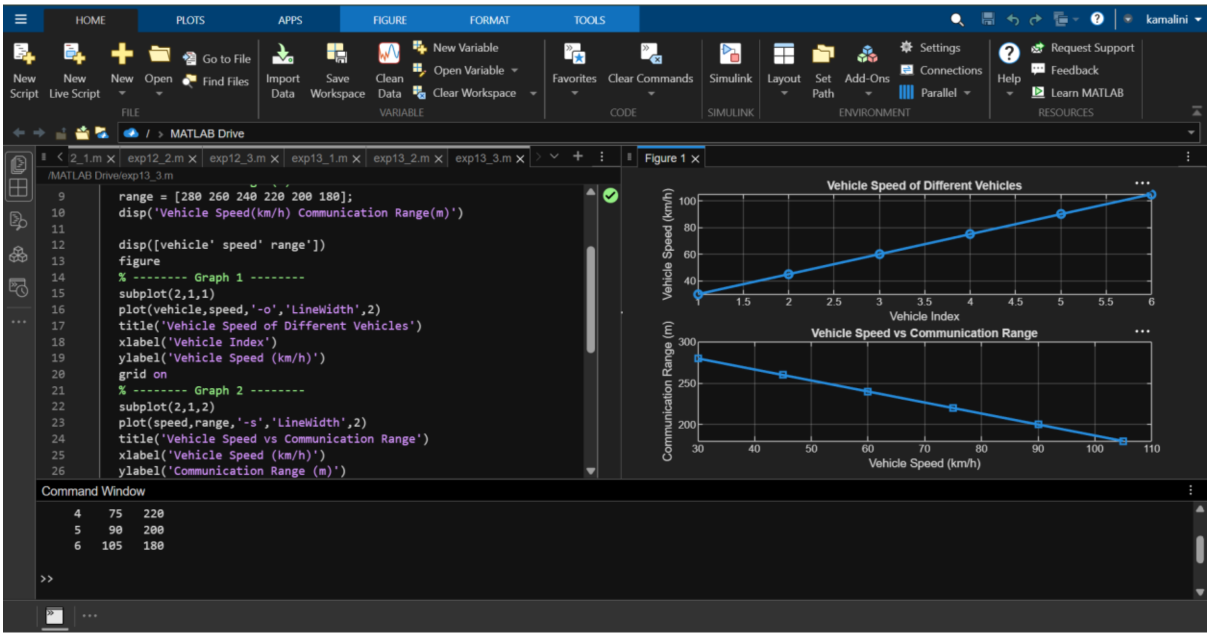
Objective 3 :

To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

CODE :

```
clc;

clear;

close all;

% Vehicle IDs
```

```
vehicle = 1:6;

% Vehicle Speed (km/h)

speed = [30 45 60 75 90 105];

% Communication Range (m)

range = [280 260 240 220 200 180];

disp('Vehicle Speed(km/h) Communication Range(m)')


disp([vehicle' speed' range'])

figure

% -------- Graph 1 --------

subplot(2,1,1)

plot(vehicle,speed,'-o','LineWidth',2)

title('Vehicle Speed of Different Vehicles')

xlabel('Vehicle Index')

ylabel('Vehicle Speed (km/h)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(speed,range,'-s','LineWidth',2)

title('Vehicle Speed vs Communication Range')

xlabel('Vehicle Speed (km/h)')

ylabel('Communication Range (m)')

grid on
```


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
kamalini
MATLAB Drive
exp12_2.m
exp12_3.m
exp13_1.m
exp13_2.m
exp13_3.m
/MATLAB Drive/exp13_3.m
range = [280 260 240 220 200 180];
disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')
ylabel('Vehicle Speed (km/h)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(speed,range,'-s','LineWidth',2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
Figure 1
Vehicle Speed of Different Vehicles
Vehicle Speed (km/h)
Vehicle Index
Vehicle Speed vs Communication Range
Communication Range (m)
Vehicle Speed (km/h)
Command Window
4 75 220
5 90 200
6 105 180
>>